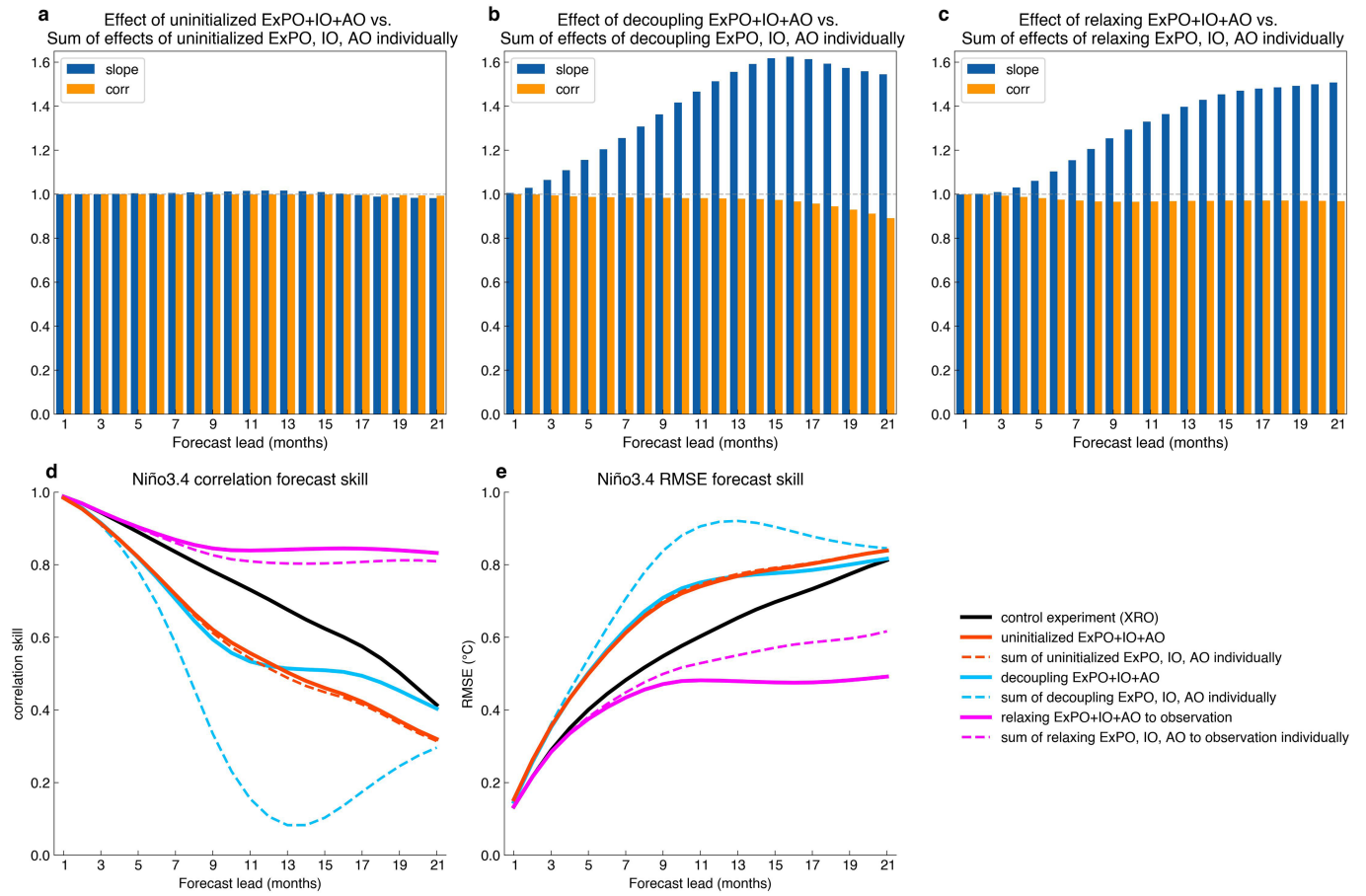




Extended Data Fig. 3 | Test of additivity (i.e., linearity) of the sensitivity experiments. **a**, Regression slope and linear correlation coefficients for the Niño3.4 SSTA forecasts between the effects of the uninitialized ExPO+IO + AO experiment ($XRO - U_{ExPO+IO+AO}$) and the sum of the effects of the individual uninitialized ExPO, IO, and AO experiments ($3 \times XRO - U_{ExPO} - U_{IO} - U_{AO}$). **b** and **c**, same as **a**, but for decoupling experiments ($XRO - D_{ExPO+IO+AO}$ vs. $3 \times XRO - D_{ExPO} - D_{IO} - D_{AO}$) and relaxing towards observation experiments ($XRO - R_{ExPO+IO+AO}$ vs. $3 \times XRO - R_{ExPO} - R_{IO} - R_{AO}$), respectively. **d**, **e** the all-months correlation skill (**d**) and RMSE (**e**) of the 3-month running mean Niño3.4 index, as a function of the forecast lead month in the control experiment (*black line*) and sensitivity experiments: the uninitialized ExPO+IO + AO experiment (*solid red line*) and sum of uninitialized ExPO, IO, and AO individually (*dashed red line*), the decoupling ExPO+IO + AO experiment (*solid blue line*) and sum of

decoupling ExPO, IO, and AO individually (*dashed blue line*), and the relaxing ExPO+IO + AO to observation experiment (*solid magenta line*) and sum of relaxing ExPO, IO, and AO to observation individually (*dashed magenta line*). The individual basin uninitialized experiments are additive with the slopes and correlations at all lead months being very close to 1. But the individual basin decoupling experiments and the individual relaxation towards observations experiments are not additive, owing to a nonlinear dependence on the operator parameters. The sum of the effects of decoupling ExPO, IO, and AO individually is much larger than the effect of decoupling ExPO+IO + AO, suggesting that the decoupling experiment framework overestimates the contribution of each basin, given the presence of indirect pathways due to interactions among basins.